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SECTION 02

Submitted To: Ma'am Asma Rauf.

**Project Title: Accident Detection and Emergency
Alarm System.**

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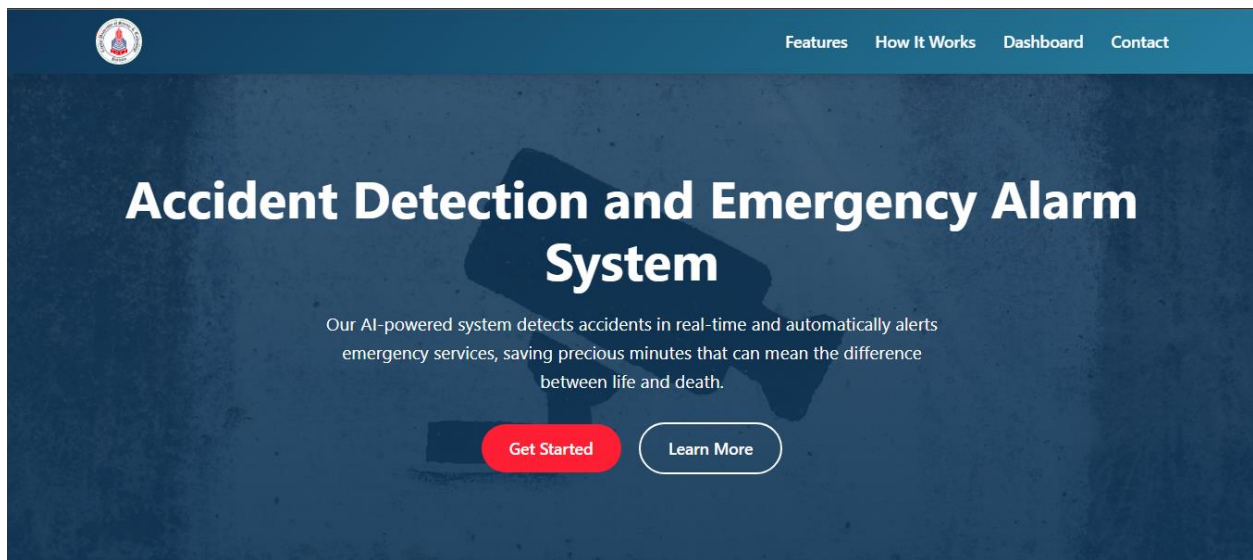
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Project Title: Accident Detection and Emergency Alarm System

Project Description:

The Accident Detection and Emergency Alarm System is developed to automatically detect road accidents and send emergency alerts to the user's trusted contacts. Using real-time sensor input from the vehicle (including accelerometers, gyroscopes, and vibration sensors), the system identifies critical incidents and validates them based on severity and duration. Once an accident is confirmed, the system records the GPS location and immediately notifies emergency contacts with the location and severity of the crash. Users can also manually trigger an SOS alert. In addition to emergency handling, the system provides features for user registration, contact management, and viewing accident history. Admins are provided with monitoring capabilities and receive alerts about system health status. This system aims to improve emergency response time and user safety in road transportation.



Functional Requirements:

Vehicle Monitoring

- FR1.1: The system shall continuously monitor the vehicle's motion using accelerometer, gyroscope, and vibration sensors.
- FR1.2: The system shall store real-time sensor data with timestamps.

Accident Detection

- FR2.1: The system shall detect accidents when sensor values exceed safety thresholds for more than 3 seconds (sustained impact).
- FR2.2: The system shall validate accidents using a rule-based algorithm (e.g., high impact followed by no movement) and set the report as validated.

Location Tracking

- FR3.1: The system shall record GPS coordinates at the time of accident detection.
- FR3.2: The system shall link the recorded location to the corresponding accident report.

Emergency Alert Generation

- FR4.1: The system shall send alerts to all emergency contacts linked to the user upon a validated accident.
- FR4.2: Alerts shall include the location and severity of the accident.

Manual SOS Feature

- FR5.1: The system shall allow users to manually trigger an SOS request using a button.
- FR5.2: The system shall create and send an alert with the user's live location.

User Management

- FR6.1: Users shall be able to register with their name, phone, email, and password.
- FR6.2: Users shall log in to manage their emergency contacts and view accident reports.

Emergency Contact Management

- FR7.1: Users shall be able to add, update, and delete emergency contacts (name, phone, relation).
- FR7.2: The system shall require at least one emergency contact per user.

Accident History

- FR8.1: The system shall maintain a log of accident reports with timestamps and severity levels.
- FR8.2: Users shall be able to filter reports by date and severity.

Admin Monitoring (Optional)

- FR9.1: Admins shall be able to view all accident reports and user accounts.
- FR9.2: Admins shall receive system health alerts, such as inactive GPS.

System Health Checks

- FR10.1: The system shall verify the status of sensors and GPS every 5 minutes and update the system status.
- FR10.2: Users shall receive notifications if the sensor or GPS status is marked as "Inactive".

Non-Functional Requirements (aligned with system scope):

The system should respond to an accident detection or manual SOS within 2 seconds.

Sensor and location data must be stored securely and be accessible only to authorized users.

Alerts should be delivered in real-time to emergency contacts via SMS, email, or push notifications.

The system interface should be intuitive and usable by people with basic mobile phone skills.

The system should remain available and operational at least 99% of the time.

Passwords and personal data should be encrypted following best practices.

The system should be scalable to accommodate multiple users and vehicles.

Sensor and GPS modules should integrate seamlessly with the main system.

The system should maintain logs of alerts, system checks, and user actions for auditing.

Admin functions should be restricted and protected through secure authentication.

Dashboard:


[Features](#)
[How It Works](#)
[Dashboard](#)
[Contact](#)

Comprehensive Dashboard

Monitor your safety status and manage emergency contacts through our intuitive dashboard.

-  **Real-time Status**
View your current protection status and system readiness at a glance.
-  **Contact Management**
Easily add or update emergency contacts who should be notified in case of an accident.
-  **Incident History**
Review past alerts and incidents with detailed timelines and response information.
-  **Customizable Settings**
Adjust sensitivity levels and notification preferences to match your needs.



Features:


[Features](#)
[How It Works](#)
[Dashboard](#)
[Contact](#)

Life-Saving Features

Our system combines cutting-edge technology with emergency response protocols to provide comprehensive protection



Instant Detection

Advanced sensors and AI algorithms detect accidents within milliseconds of impact, analyzing multiple parameters for accuracy.



Automatic Alerts

Immediately notifies emergency services with precise location and accident details, even if you're unconscious.



GPS Tracking

Provides exact coordinates to first responders, reducing response time in critical situations.



Vital Monitoring

Optional wearable integration monitors heart rate and other vital signs to assess injury severity.



Emergency Contacts

Automatically alerts your designated emergency contacts with your location and status.



Accident Analytics

Comprehensive reporting helps identify accident patterns and improve safety measures.

How It Works:

[Features](#)[How It Works](#)[Dashboard](#)[Contact](#)

How SafeGuard Works

Our system operates seamlessly in the background, ready to spring into action when needed most

- 1 Continuous Monitoring**

Our system constantly analyzes data from vehicle sensors, including impact force, sudden deceleration, airbag deployment, and unusual vehicle orientation.
- 2 Accident Detection**

Advanced machine learning algorithms evaluate the sensor data to distinguish between normal driving events and actual accidents with 98.7% accuracy.

[Features](#)[How It Works](#)[Dashboard](#)[Contact](#)

- 3 Emergency Alert**

When an accident is detected, the system immediately establishes a connection with emergency services, transmitting GPS coordinates, time of impact, and severity assessment.
- 4 Two-Way Communication**

If possible, the system opens a voice channel with emergency responders and plays pre-recorded information while waiting for human response.
- 5 Post-Accident Support**

After the incident, the system provides accident details to insurance companies and helps coordinate follow-up care if needed.

Contact Page:



Get SafeGuard Today

Contact us to learn more about protecting yourself and your loved ones

Full Name

Email Address

Phone Number

I'm interested in

Personal Protection 

Message

Send Message



Advanced accident detection and emergency response technology designed to save lives when seconds count.



Quick Links

[Features](#)
[How It Works](#)
[Dashboard](#)
[Contact](#)

Support

[FAQ](#)
[Installation Guide](#)
[Troubleshooting](#)
[Privacy Policy](#)
[Terms of Service](#)

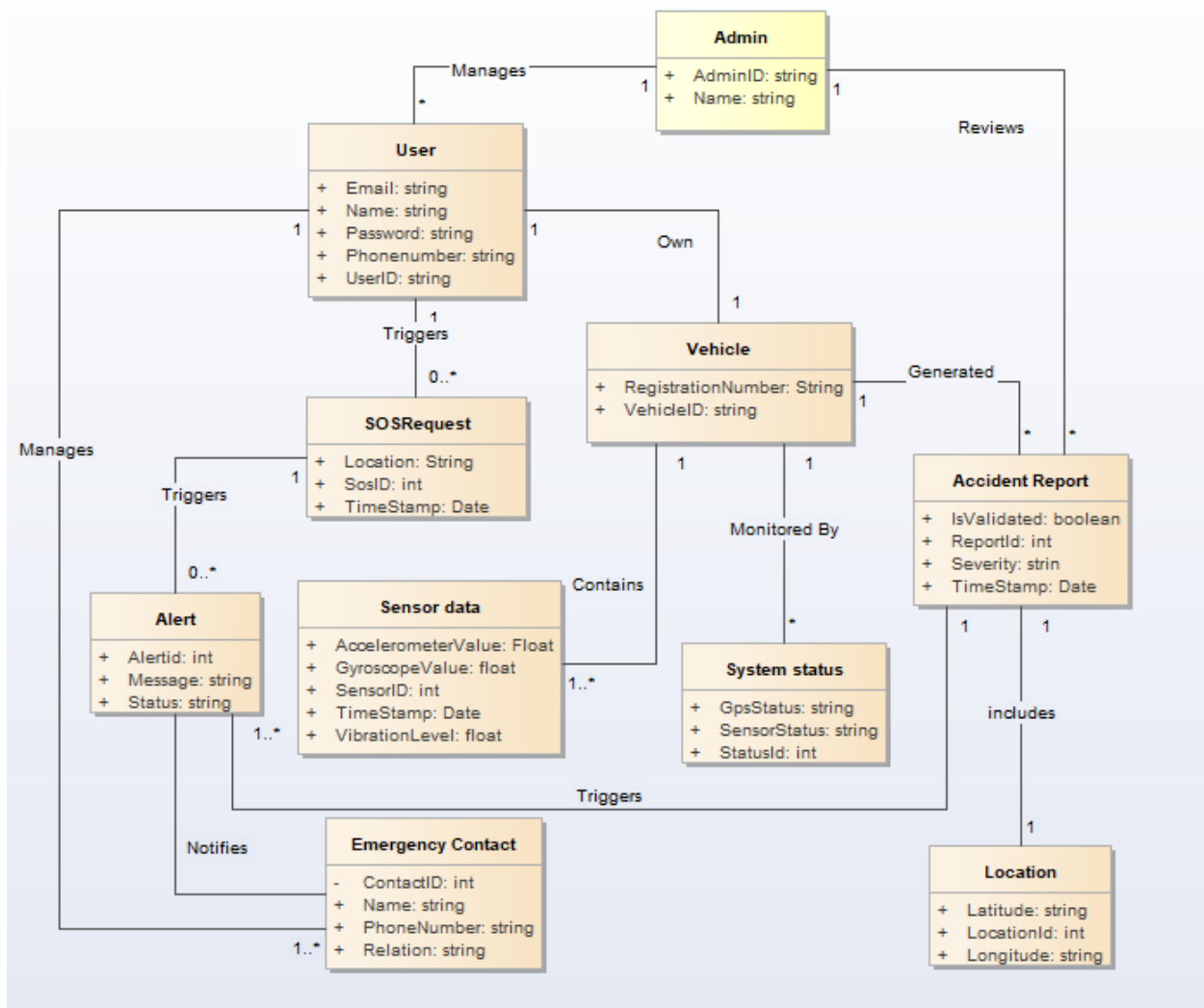
Newsletter

Subscribe to receive updates, safety tips, and special offers.

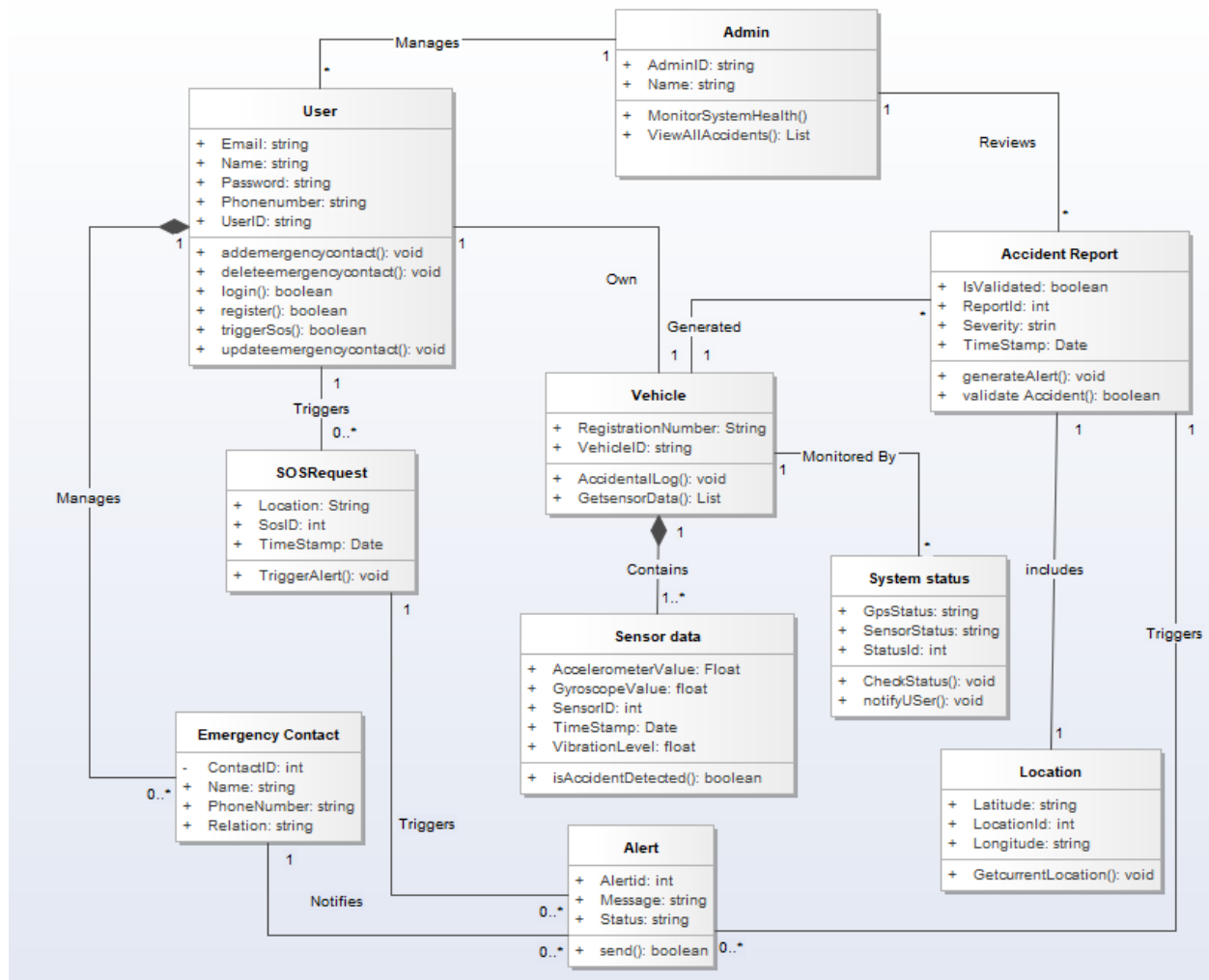
Your email address



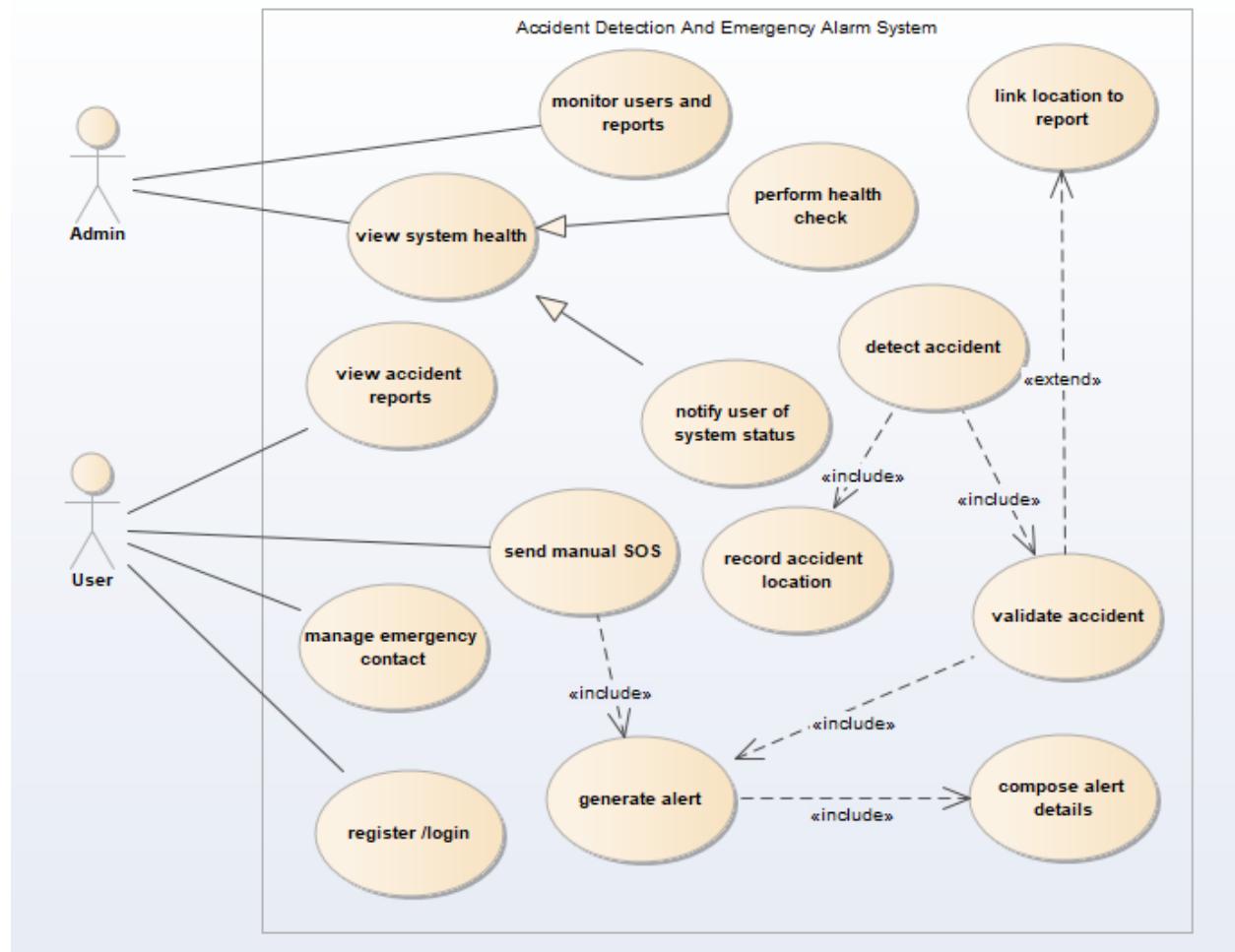
Domain Model:



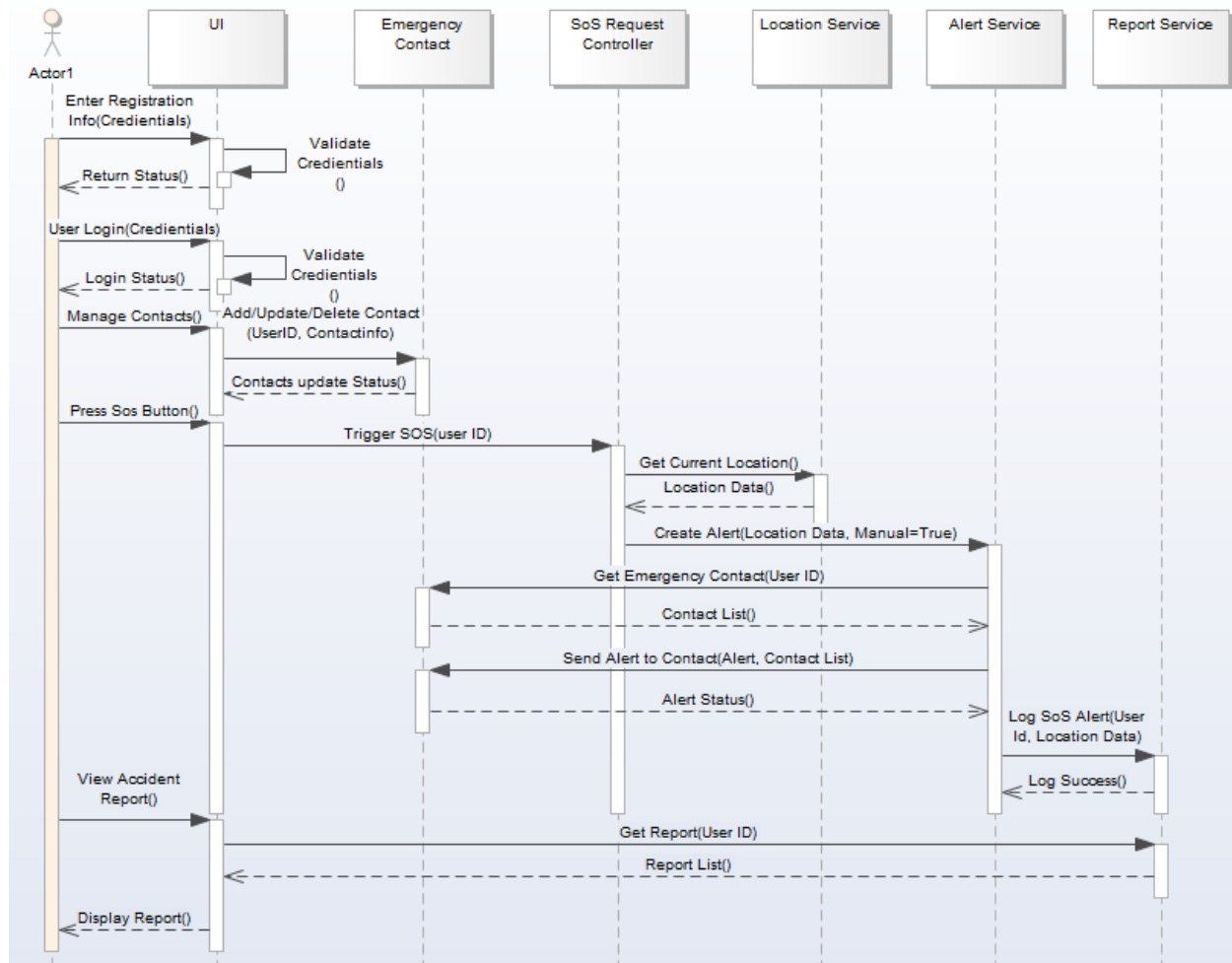
Class Diagram:



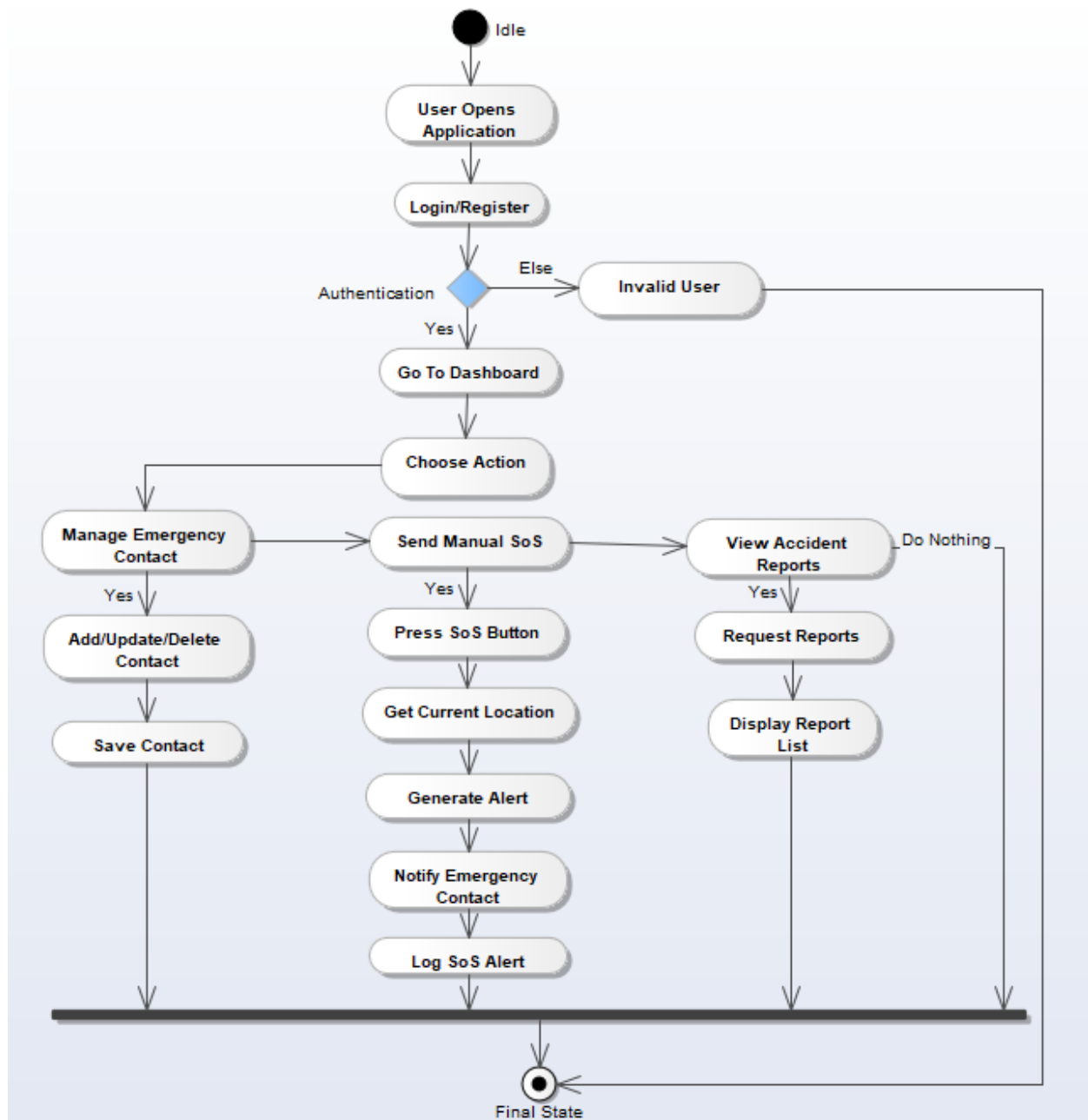
Use Case Diagram:



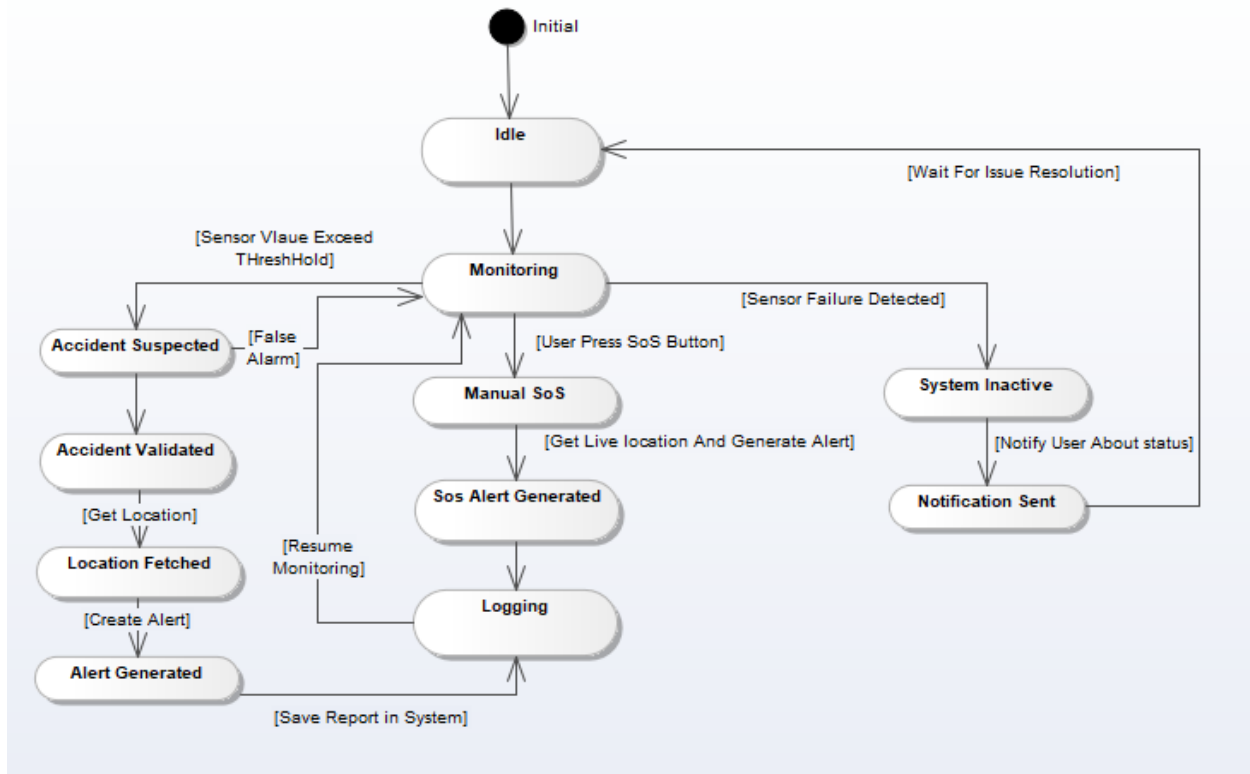
Sequence Diagrams:



Activity Diagram:

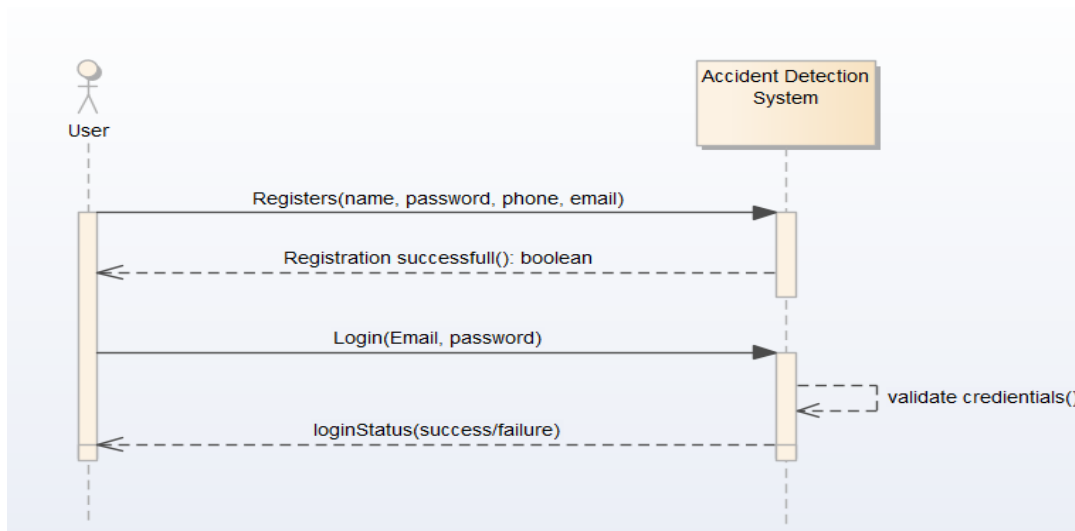


State Machine diagram:

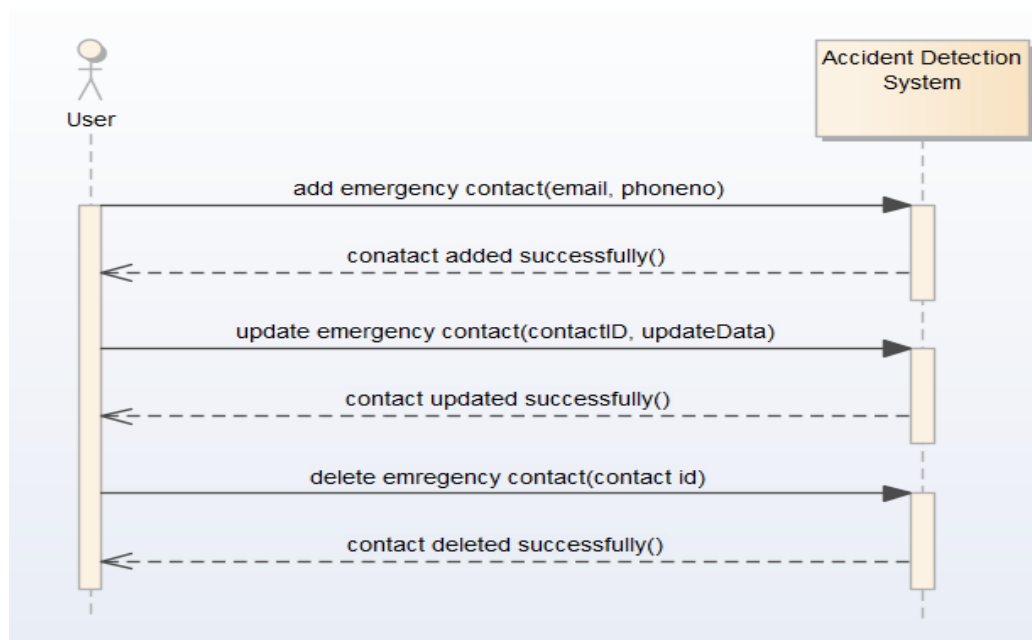


System Sequence Diagrams:

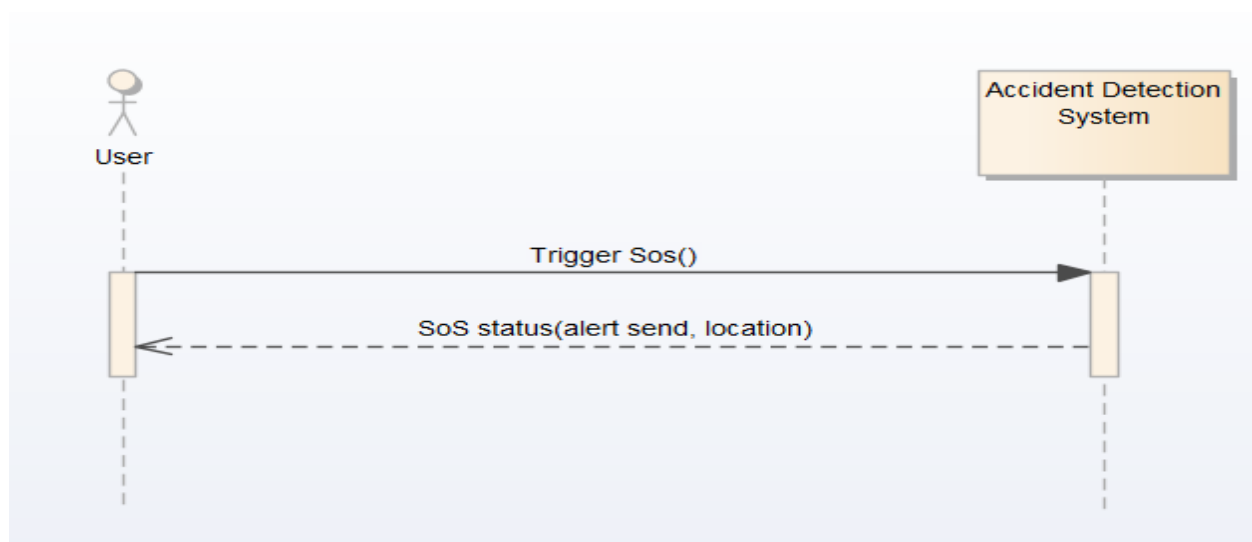
1-User registration and login:



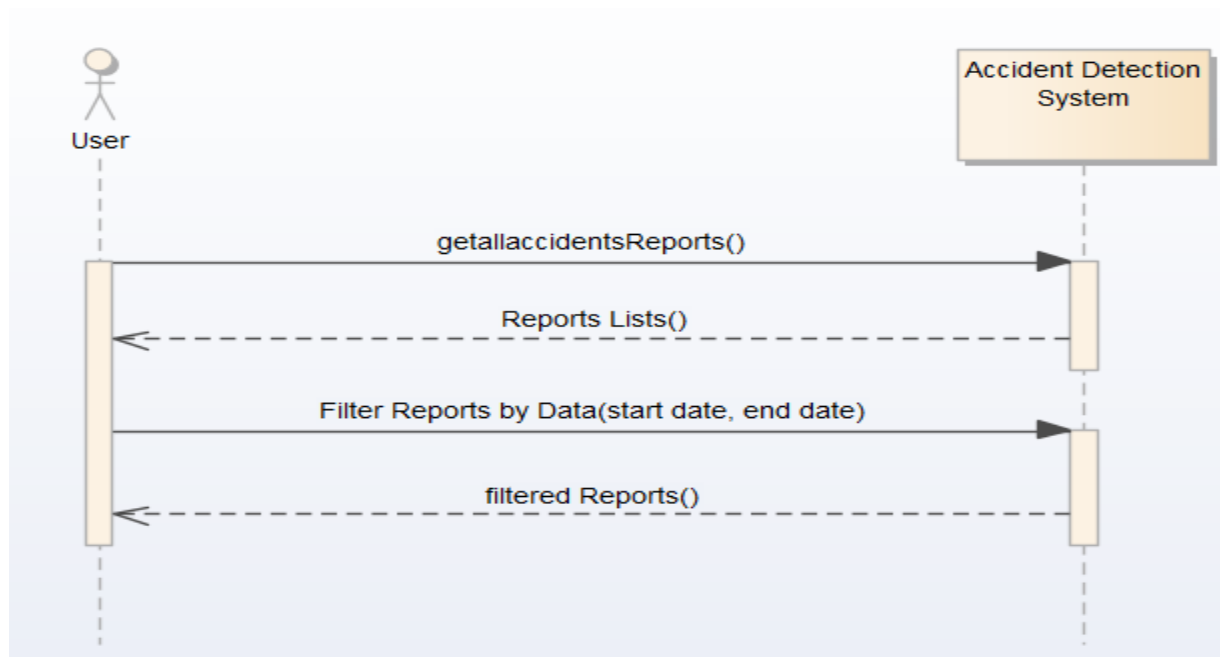
2-Manage emergency contacts:



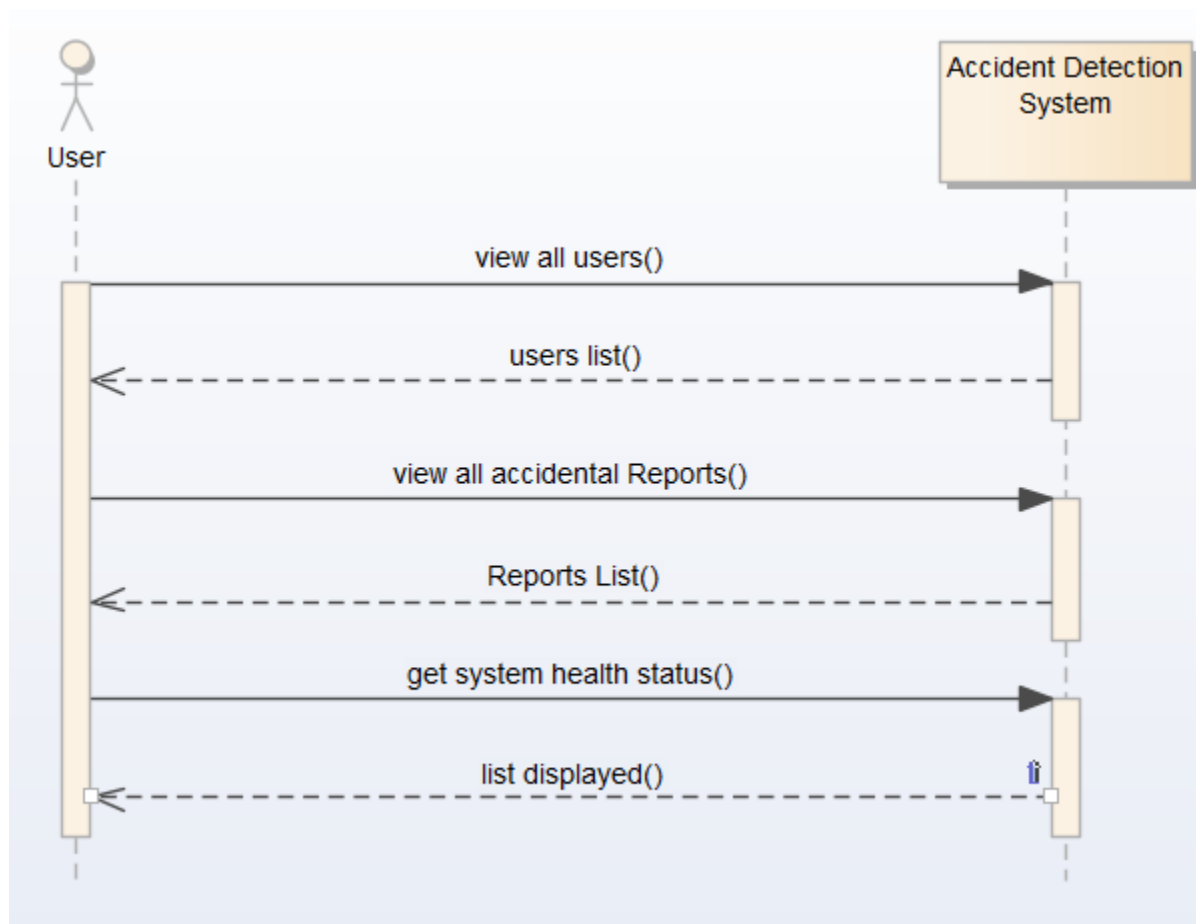
3-Send Manual SOS:



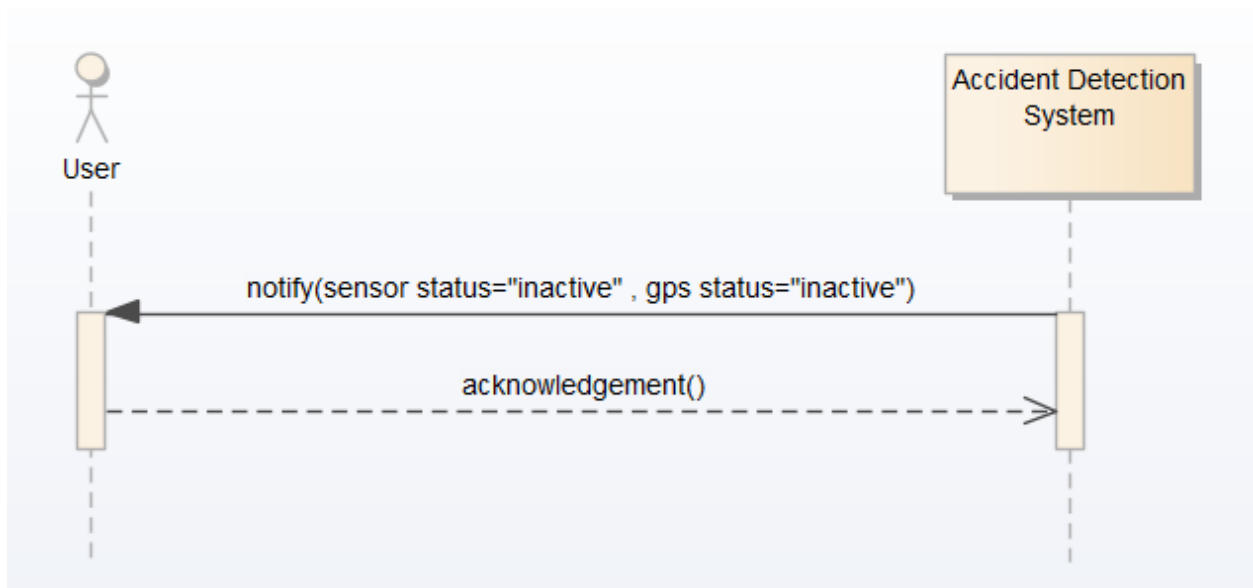
4-View and filter accident Reports:



5-Admin Monitoring and Alert:



6-System Health status to users



7-Accident detection flow:

